Subject Code	**AF5365**
Subject Title	**Applications of Computing and Technology in Accounting and Finance I**
Credit Value	3
Level	5
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	None
Objectives	This subject is designed to study the scientific computing skills and apply the skills in accounting and finance. The subject covers basic stochastic modeling, uses Python/R/VBA to value different financial products and do static/dynamic risk hedging and cash flow replications using Monte Carlo method, Variance Reduction method, Metropolis Hasting and Gibbs sampler, and other methods. Direct Market Access is introduced with the applications in electronic trading using scientific computing tools. After studying this subject, the student should master the necessary analytical tools for further study and work. This subject contributes to the achievement of the MSc in Accounting and Finance Analytics programme learning outcomes by enabling students to apply technology and data analytics skills to solve accounting and finance problems faced in real-life situations in an ethical manner (Outcome 3).
Intended Learning Outcomes	Upon successful completion of this course, students should be able to: a. Master the basic scientific computing skills b. Achieve the direct market access c. Understand the risk measures and their calculation using the real data d. Utilize the Monte Carlo methods to simulate the financial products' dynamics and implementing pricing models of derivatives
Subject Synopsis/ Indicative Syllabus	**Business Ethics** The issues in applying business technology and analytical tools in an ethical way. **Basic Scientific Computing** The issues in the use of scientific computing software packages such as Python/R/VBA. **Monte Carlo Methods (including Derivative Pricing)** How to implement the Monte Carlo methods to simulate the financial products' dynamics, valuate the derivative products' prices and formulate corresponding static / dynamic hedging strategies; scenario analysis in financial statements and stress test. **Electronic Trading / Direct Market Access (DMA)** Utilize the Application Programming Interface (API) to achieve the direct market access; understand the basic order types and how to construct the electronic trading platforms. **Risk Measures** Introduce the risk measures and know how to calculate the measures using scientific computing based on the data from DMA.

Teaching/Learning Methodology	Key concepts and techniques will be introduced through lectures. The subject places a lot of emphasis on project work. Students will be required to deliver a project which emphasizes on real-world accounting and finance issues. By completing the project, students should have hands-on experience in using the knowledge they have learnt in class to solve accounting and finance problems in practice. Students are encouraged to share their views and experiences actively with their lecturer and classmates.
Assessment Methods in Alignment with Intended Learning Outcomes	(see below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
		a	b	c	d
1. Class participation	5%	√	√	√	√
2. Quizzes & assignment	25%	√	√	√	√
3. Group Project	20%	√	√		
4. Final examination	50%	√	√	√	√
Total	100 %				

Class participation – Students have to read assigned reading materials and complete exercises to participate actively in class discussion, which would assess their understanding of the key concepts and techniques, and their applications.

Quizzes & assignment – are used to test students' ability in understanding the materials and in achieving the intended learning outcomes through a more in-depth investigation of different topics.

Group Project – Group project will be used for students to apply what is taught in class and allow the students to learn from one another.

Final Examination – Final examination are used to test students' overall ability in applying the knowledge learnt in the subject.

Note: The specific requirements on individual assessment components discussed above could be adjusted based on the pedagogical needs of subject lecturers.

Student Study Effort Expected		
	Class contact:	
	▪ Lectures / Seminars	39 Hrs.
	Other student study effort:	
	▪ Reading materials / textbook questions	39 Hrs.

	▪ On average around 16 hours will be spent on the individual critique and around 20 hours for the group project discussion, presentation and written report	36 Hrs.
	Total student study effort	114 Hrs.
Reading List and References	***References*** Monte Carlo Methods in Financial Engineering, 2003 Edition, P. Glasserman, Springer Algorithmic Trading and DMA: An introduction to direct access trading strategies, 2010 Edition, Barry Johnson, 4Myeloma Press.	